Slide 1: Executive Summary & Corporate Profile

Corporate Identity & Strategic Focus

Company: Sagar Fresh Technologies Pvt. Ltd. (Bengaluru, India)

Sector: Direct-to-Consumer (D2C) Fresh Seafood, Prime Meats & Cold-Chain Logistics

Core Value Proposition: Farm-to-fork fresh coastal fish and pasture-raised meats delivered within 90 minutes inside reusable, IoT temperature-monitored cold boxes.

Primary Technological Moat: Circular FreshSeal container logistics with automated refundable deposits and real-time probe telemetry reporting (0°C to 4°C).

Executive Thesis

"India's organized online meat commerce currently generates over 12,000 tonnes of single-use thermocol and plastic waste every year while suffering cold-chain micro-breaches during city traffic. **Sagar Fresh** replaces disposable packaging with a reusable cold-gel vault model, guaranteeing 100% lab-certified cold-chain compliance and saving up to ₹39 per order in operating packaging expenditure."

90 Mins	0°C - 4°C	96.0%	₹1,049
HYPERLOCAL DELIVERY SLA	GUARANTEED FSSAI SAFE ZONE	COOLER BOX RETURN RATE	AVERAGE ORDER BASKET (AOV)

Slide 2: Critical Industry Challenges

1. Cold-Chain Temperature Breaches

- Traditional wet markets operate at ambient temperatures (28°C–34°C), accelerating bacterial growth and moisture deterioration within 45 minutes of inward.
- Existing online delivery platforms use standard cardboard boxes with small ice gel packs that thaw during metro traffic delays (65–85 minutes).
- Core package temperatures routinely spike above 7°C, compromising freshness without customer visibility.

2. Packaging Cost & Waste Drain

- Market incumbents spend between ₹38 and ₹55 per delivery on single-use thermocol linings, foam sheets, and bubble wrap.
- This cost represents dead capital burned on every fulfillment cycle.
- Creates substantial municipal landfill waste, alienating environmentally conscious urban millennial and Gen-Z households.

3. Zero Verification & Trust Deficit

- Current brands market products as "chilled, never frozen" but provide zero proof to the consumer upon arrival.
- Consumers frequently experience water-logged meat from thawed ice or chemical odor from formalin-treated seafood.
- High return rates and customer churn due to inconsistent freshness across seasonal heatwaves.

Economic Impact of Current Industry Inefficiencies

Incumbents suffer an estimated **4.8% customer refund rate** due to temperature degradation in transit and spend over **7.2% of net revenue** exclusively on non-recoverable disposable packaging materials.

Slide 3: The Sagar Fresh Solution — FreshSeal™

1. Precision Harbor-Side Inward & Blast Chilling

Daily catches inward from Mangalore and Ratnagiri jetties by 4:00 AM. Products are descaled, washed in RO-purified water, portioned into neat cuts, and immediately blast-chilled to 0°C–2°C within 30 minutes, preserving cellular muscle texture.

2. ThermalVault™ Reusable Cold-Gel Boxes

Constructed from high-density Expanded Polypropylene (EPP) with food-grade phase-change gel plates. Maintains core temperature below 4°C for up to 36 continuous hours without external power.

3. Real-Time Consumer IoT Telemetry

Each container features an integrated digital thermal sensor. Customers track live internal box temperatures on their web application in real time during transit (0°C to 4°C safety gauge).

4. Automated Deposit & Return Engine

A refundable deposit of ₹250 is included at checkout. Upon scheduled collection of the empty container by return couriers, ₹250 is instantly credited back to the customer's UPI account via automated webhook.

Operating Workflow Diagram

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[Harbor Jetty 4:00 AM] ➔ [Central Cold Hub (0°C-2°C)] ➔ [ThermalVault Box Packed] ➔ [Live IoT Tracking 90 Min Transit] ➔ [Delivery] ➔ [Instant UPI Deposit Return]
```

Slide 4: Market Opportunity & Growth Vectors

TOTAL ADDRESSABLE MARKET (TAM)

$45 Billion

Total Indian retail meat and seafood commerce (currently 92% unorganized wet market vendors).

SERVICEABLE ADDRESSABLE MARKET (SAM)

$4.8 Billion

Organized meat and fresh fish consumption across Tier-1 urban metropolises in India.

SERVICEABLE OBTAINABLE MARKET (SOM)

$42 Million

3-Year achievable target capturing 3.5% share across high-density pin codes in Bengaluru and Mumbai.

Underlying Market Tailwinds Supporting Sagar Fresh

FSSAI Regulatory Crackdown: Increased state inspections prohibiting formalin and chemical preservation in traditional fish landings.

Disposable Income Growth: Upper-middle-class urban families actively switching to certified antibiotic-free meat with a 15–20% willingness to pay premium.

Zero-Waste Consumer Preference: Over 70% of premium urban households prefer sustainable, circular packaging over disposable thermocol.

Slide 5: Product Catalog & Supply Chain

PRODUCT NAME	CATEGORY	SOURCING REGION	PORTION & NET WEIGHT	COLD-CHAIN STANDARD	PRICE
Seer Fish / King Mackerel Steaks	Coastal Finfish	Mangalore Harbor Jetty, Karnataka	4-6 Steaks (Net 450g / Gross 650g)	0°C to 2°C Blast Chilled	**₹799**
White Pomfret Whole Cleaned	Coastal Finfish	Ratnagiri Jetty, Maharashtra	2 Whole Fish (Net 380g / Gross 500g)	Degilled & Diamond Scored	**₹689**
Sea Tiger Prawns (Jumbo Deveined)	Crustaceans	Visakhapatnam Deep Sea, Andhra Pradesh	14-18 Pieces (Net 300g / Gross 500g)	Peeled, Deveined, Tail-on	**₹599**
Rich Tender Goat (Curry & Biryani Cut)	Pasture Mutton	Mandya & Bannur Co-ops, Karnataka	Medium Bone-In (Net 500g)	Grass-fed under 9 months, RO Washed	**₹629**
Classic Chicken Breast Fillets	Fresh Poultry	Coimbatore Biosecure Farms	2-3 Fillets (Net 500g)	100% Antibiotic & Hormone-Free	**₹289**
Free-Range Country Chicken	Heritage Poultry	Chittoor Free-Range Co-op	Curry Cut (Net 550g / Gross 700g)	Desi breed, yellow fat retained	**₹389**
Amritsari Fish Tikka (Ready to Cook)	Marinated Cuts	Sagar Fresh Master Kitchen	8-10 Cubes (Net 350g)	Chef-marinated in mustard oil & spices	**₹369**

Slide 6: Reusable Container Economics

ThermalVault Unit Cost Structure

Manufacturing Cost per EPP Thermal Box	**₹420.00**
Certified Useful Life Cycle	**150 Deliveries**
Amortized Box Depreciation per Delivery	**₹2.80**
Ultrasonic Cleaning & Sanitization per Cycle	**₹6.00**
Net Packaging Cost per Sagar Fresh Delivery	**₹8.80**

Incumbent Disposable Packaging Cost Comparison

Corrugated Exterior Box with Lamination	₹18.00
EPS Thermocol Insulated Inner Liner	₹16.00
Single-Use Ice Gel Packs (2 Units)	₹10.00
Multi-layer Tape & Secondary Bubble Wrap	₹4.00
Total Disposable Packaging Cost (Incumbents)	**₹48.00**

₹39.20
PACKAGING SAVINGS PER ORDER

150x
CONTAINER REUSABILITY FACTOR

10 Mins
AVERAGE INSTANT UPI REFUND TAT

Slide 7: Competitive Landscape Analysis

FEATURE / CAPABILITY	SAGAR FRESH	LICIOUS	FRESHTOHOME	LOCAL WET MARKETS
Delivery Speed	**90-Minute Express**	90-Minute / Scheduled	Next-Day / 90-Minute	Immediate (In-Store)
Cold-Chain Telemetry	**Live IoT Probe (0°C–4°C)**	**Unverified Marketing Claim**	**Unverified Marketing Claim**	**No Temperature Control**
Packaging Material	**100% Reusable Thermal Vault**	Disposable Thermocol / Pouch	Cardboard & Plastic Liners	Single-Use Plastic Bags
Packaging Cost / Order	**₹8.80 (Circular Amortized)**	₹48.00–₹55.00	₹42.00–₹50.00	₹2.00 (Unregulated)
Customer Retention Mechanism	**₹250 Deposit Float / Credit**	Membership Subscription	Wallet Cashback Points	Personal Habit
Food Safety & Testing	**RO-Washed & Lab Certified**	Central Lab Certified	Chemical-Free Claim	**High Formalin Contamination Risk**

Slide 8: Unit Economics & Contribution Margins

Order Economics Breakdown (₹ INR)

Gross Merchandise Value (AOV)	**₹799.00**	**100.0%**
Cost of Goods Sold (Farmgate Procurement & Cleaning)	₹471.40	59.0%
Gross Profit	**₹327.60**	**41.0%**
Cold Box Amortization & Sanitization (Circular)	₹8.80	1.1%
Last-Mile EV Delivery Payout	₹55.00	6.9%
Payment Gateway Fees (Blended UPI / Cards)	₹8.00	1.0%
Central Cold Hub Warehousing & Energy	₹18.00	2.3%
Contribution Margin 1 (CM1)	**₹237.80**	**29.7%**
Customer Acquisition Cost (Blended CAC)	₹120.00	15.0%
Contribution Margin 2 (CM2 - Net Operational Profit)	**₹117.80**	**14.7%**

Unit Margin Profitability Takeaways

- **First-Order Positive:** Unlike traditional D2C competitors that lose money on the initial customer purchase, Sagar Fresh generates a positive CM2 of ₹117.80 on Order #1.
- **Sustainable Packaging Advantage:** Lower packaging overhead (+₹39.20 per basket) offsets higher quality blast-chilling investments.
- **LTV / CAC Expansion:** Repeat orders experience an acquisition cost of ₹0.00, lifting CM2 on repeat purchases to **₹237.80 (29.7% net margin)**.

Slide 9: Go-To-Market & Customer Acquisition

1. Hyperlocal Micro-Market Launch

Focus initial operations strictly on high-density Bengaluru gated residential complexes (Mahadevapura, Indiranagar, Whitefield, Koramangala).

Gated community security receptions allow batch drop-off and morning pickup routines, drastically cutting delivery partner idle time.

2. Digital Advertising Channels

Meta Ads (Instagram & Facebook): Short-form video highlighting harbor inward at 4:00 AM vs. warm supermarket displays.

Google Paid Search: Capturing high-intent queries: "fresh surmai delivery bangalore", "fresh mutton 90 mins", "licious alternative".

3. Retention & Wallet Incentive

When scheduling container pickups, customers are offered an instant choice:

- ₹250 Refund to Bank via UPI
- **₹300 Sagar Fresh Wallet Credits** (+₹50 Bonus for next order)

Drives 42% wallet retention, locking in repeat purchasing cycles within 8 to 10 days.

₹120	3.8x	4.2 Orders	42%
TARGET BLENDED CAC	TARGET 6-MONTH LTV / CAC	MONTHLY REPEAT FREQUENCY	WALLET RE-INVESTMENT RATE

Slide 10: Technical Architecture & IoT Systems

Technology Stack Overview

- **Frontend:** React 19 Single Page Application with Tailwind CSS v4, dynamic catalog drawers, and mobile-responsive viewport.
- **Backend API:** Node.js & Express 4 RESTful architecture with secure JSON Web Token (JWT) authentication and role-based access control.
- **Cloud Database:** MongoDB Atlas cloud cluster with Mongoose ODM schemas for users, products, orders, coolers, and return tracking.
- **Cloud Deployment:** Vercel Serverless environment with automatic edge asset distribution and dynamic API rewrites.

Data Entities & IoT Monitoring

- **Container Fleet Telemetry:** Each cooler record stores RFID serials, battery charge, sanitation logs, and lifetime cycle wear.
- **Live Probe Engine:** Ingests periodic temperature measurements into the order document, generating real-time audit graphs on consumer devices.
- **Automated Refund Webhook:** Triggers deposit balance updates and generates UPI transaction receipts upon operational collection verification.

Cloud Deployment Flow

```
GitHub (Dhinesh1628/sagarfresh) → Automated Vercel Build → Serverless Node Runtime → MongoDB Atlas Cloud (Sharded Cluster)
```

Slide 11: Current Traction & MVP Verification

100%	99.6%	1.8°C	0 Crashes
AUTOMATED TEST PASS RATE	THERMAL COMPLIANCE METRIC	AVERAGE IN-TRANSIT TEMP	FAIL-SAFE DATABASE RESILIENCE

Verified Functional Workflows

- **Credential Authentication:** Full role separation between operations administration and consumer accounts.
- **Dynamic Cart & Deposit Engine:** Real-time checkout calculations incorporating the ₹250 refundable container deposit.
- **Live Telemetry Simulation:** Customer order tracking displaying active temperature probe telemetry between 0°C and 4°C.
- **Return & Refund Loop:** Instant cooler pickup scheduling with UPI refund verification.

Database Integrity Verification

- Live cloud synchronization verified on **MongoDB Atlas** (Cluster shard `ac-6frwnfu-shard-00-02`).
- 8 Indian coastal and livestock SKUs populated with net/gross weights, portion counts, and origin jetty tracking.
- Fleet inventory initialized across active delivery, customer custody, and sanitization depot states.

Slide 12: Financial Forecasts & Growth Roadmap

GROWTH INDICATOR	MONTH 6 (PILOT HUB)	MONTH 12 (CITY SCALE)	MONTH 18 (MULTI-CITY EXPANSION)
Operating Central Cold Hubs	1 Hub (Bengaluru East)	3 Hubs (Bengaluru Wide)	6 Hubs (Bengaluru + Mumbai)
Monthly Completed Orders	4,500 Orders	22,000 Orders	65,000 Orders
Monthly Gross Revenue (₹ INR)	₹35.9 Lakhs ($43K)	₹1.75 Crores ($210K)	₹5.19 Crores ($625K)
Active Cooler Containers in Fleet	800 Units	3,200 Units	8,500 Units
Container Recovery Rate	94.5%	96.2%	97.0%
Contribution Margin 2 (Net Operating Profit)	**₹4.8 Lakhs (13.3%)**	**₹28.1 Lakhs (16.0%)**	**₹98.6 Lakhs (19.0%)**

Strategic Execution Milestones

Phase 1 (Months 1–6): Optimize Mahadevapura hub, establish 96% return rate across 25 gated societies, reach positive unit economics.

Phase 2 (Months 7–12): Expand to 3 micro-hubs across Bengaluru, integrate Bluetooth Low Energy physical loggers, supply high-end restaurants.

Phase 3 (Months 13–18): Enter Mumbai market (Bandra & Andheri hubs), partner with commercial EV fleet logistics for dedicated delivery routes.